

Textbooks

- Primary Textbooks

- 1) Engineering Circuit Analysis (ECA)

- 8th Edition, Tata McGraw-Hill
- By W.H. Hyatt, J.E. Kimmerley, and S.M. Durbin

- 2) Signals and Systems (SAS)

- 2nd Edition, Prentice Hall
- By Oppenheim, Willsky and Nawab

Class	Topics	Book sections	Practice problems
8 th Aug	Introduction to the course	(ECA) Chapter 1, Appendix 5	(SAS) 1.48, 1.49, 1.50, 1.52, 1.56
10 th Aug	Vectors, dot product and orthogonality of signals, Fourier series representation	(ECA) 18.1	(ECA) 18.1 [1,2]
12 th Aug	Fourier series analysis and synthesis, examples	(ECA) 18.1, 18.2	(ECA) 18.1 [3,4,6]
14 th Aug	No class	—	—
17 th Aug	Square wave FS, symmetry properties, alternate representations	(ECA) 18.4 (SAS) 3.1, 3.3	(ECA) 18 [14,20] (SAS) 1.32, 3.3, 3.5
19 th Aug	Complex FS representation, orthogonality, Fourier spectrum	(SAS) 3.1, 3.3	(SAS) 3.4, 3.21, 3.22(b,c), 3.23(c,d)
22 nd Aug	FS properties & convergence	(SAS) 3.4, 3.5	(SAS) 3.7, 3.40, 3.44, 3.47

Class	Topics	Book sections	Practice problems
24 th Aug	Systems, basic examples, unit impulse signal, unit step signal	(SAS) 1.5 (continuous time), 1.4.2	(SAS) 1.13, 1.38
26 th Aug	Unit impulse & unit step signals, system properties	(SAS) 1.4.2, 1.6	—
29 th Aug	Quiz – 1	—	—
31 st Aug	LTI systems, impulse response	(SAS) 1.6, 2.2	(SAS) 1.27, 1.31, 2.43(a)
5 th Sep	LTI systems, convolution operator and properties, system interconnections, example	(SAS) 2.2, 2.3, 1.5.2	(SAS) 2.8, 2.11, 2.12, 2.22(c)
7 th Sep	Stability of LTI systems	(SAS) 1.6, 2.3	(SAS) 2.14, 2.23, 2.48(a)
9 th Sep	Causality of LTI systems, Laplace transform	(SAS) 1.6, 9.1	(SAS) 1.17, 2.48(e,f), 9.1

Class	Topics	Book sections	Practice problems
12 th Sep	Laplace transform examples	(SAS) 9.1, 9.2	(SAS) 9.2, 9.3, 9.4
14 th Sep	Laplace transform – poles and zeros, inverse Laplace transform	(SAS) 9.1, 9.2, 9.3	(SAS) 9.7, 9.21 (a,b,c)
16 th Sep	Laplace transform properties	(SAS) 9.3, 9.5, 9.6	(SAS) 9.21 (d,g,i), 9.29, 9.41
21 st Sep	Mid Exam (8:30 – 10 AM)	—	—
26 th Sep	Laplace transform – causality & stability of LTI systems, RC circuit	(SAS) 9.7	(SAS) 9.9, 9.15, 9.18(a), 9.22(a,b,c), 9.26, 9.28, 9.32, 3.19 & 3.20
31 st Oct	Laplace transform for circuit analysis	(SAS) 9.7.3 (ECA) Chapter 14	(SAS) 9.64(a) (ECA) Ch14[61, 63, 64]
1 st Dec	Final Exam (3:00 – 6:00 PM)	—	—